

Microeconomic Analysis 1

A Gentle Introduction

Mark Dean
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Columbia University

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Plan

- 1.Introduction (to the course)
- 2.Introduction (to the first topic)

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Intro to the Course: What?

- 'Choice Theory'
 - Almost all economic models have, at their heart, a model of individual behavior
 - Ever since the Lucas critique
 - These models therefore underlie (most) economic analysis
 - Almost always, these models assume the agent is 'rational'
 - i.e maximizes a well specified utility function relative to some constraints
 - The aim of this part of the microeconomics course is to study the properties of these models
 - Testable implications
 - Relationship between parameters and behavior
 - Later micro courses will think about what happens when these agents interact
 - Equilibrium, game theory, etc

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Intro to the Course: What?

- Five main topics
 1. Choice, Utility and Preferences (c. 4 lectures)
 2. Consumer Theory (c. 2 lectures)
 3. Producer Theory (c. 2 lectures)
 4. Choice under Risk and Uncertainty (c. 3 lectures)
 5. Choices over Time (c. 2 lectures)

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Intro to the Course: Why?

- There are four main reasons to take this course
 - Other than the fact you have to
- 1. Some of you will end up doing research in related areas, and this is your introduction
 - Consumer theory, decision theory, industrial organization, behavioral economics etc.
- 2. Almost all of you will end up using the models that we will learn about in this class
 - Worth spending some time understanding their properties, strengths, weaknesses etc
- 3. You will use what you learn in this class in others in your first year
- 4. Introduction to the type of rigorous thinking required by economists
 - Or at least it was for me!

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Intro to the Course: Where, When, etc?

- See syllabus!

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Intro to the Course 1: How?

- A lot of the course will be quite standard
 - Introduction to choice theory, consumer theory and producer theory
 - Using mathematical tools to prove results which show what these models imply for behavior
- However, I *am* a behavioral economists, and some of this will sneak into the course
 - Use data from experimental economics to apply some of the tests we come up with
 - Discuss some of the descriptive failures of the standard models, and the alternatives that behavioral economists have come up with

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Intro to the Course 1: A note about pacing

- An inescapable fact about the start of the first year program is that you will have very heterogeneous levels of prior experience
- This means that parts of the course will always be too fast for some and too slow for others
- If you are in the former camp, DON'T PANIC!
 - You are not alone
 - there are lots of opportunities for you to get help
 - Make sure you use them, and use them early!
 - My aim is to get all of you through the 1st year
- If you are in the latter camp, don't worry!
 - Very few people come out of the 1st year feeling like it was too easy
 - Enjoy the fact that this course is not taxing you too much
 - Feel free to look slightly smug

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Introduction to the First Topic

- In the first 4 lectures or so we are going to talk about the relationship between
 - Two fundamental models of economic behavior
 - Utility maximization
 - Preference maximization
 - And the data they are designed to explain
 - Choices
- What I want to get across in this introduction is an idea of why there is anything of interest here
 - i.e. why are we going to have to study this for 4 lectures?
 - Surely utility maximization is fairly straightforward?
- This introduction is going to be very 'light'
 - So relax!

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Utility Maximization

- The model of utility maximization is probably the most pervasive in all of economics
- I am sure you have come across it
- The question I want to ask today is: how can we test it?
 - i.e. if I observe someone's behavior, how can I tell if they are in fact a utility maximizer
 - Equivalently, what predictions about behavior does the model of utility maximization make?

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Testing Utility Maximization

- In order to understand how to test the model of utility maximization (or indeed any model) we need two elements
 1. The data we are going to use
 2. A precise description of the model

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The Data

- We observe:
 - The **choices** someone makes
 - What they were choosing **from**
- Example: choices from different sets of snack foods

Available Snacks	Chosen Snack
Jaffa Cakes, Kit Kat	Jaffa Cakes
Kit Kat, Lays	Kit Kat
Lays, Jaffa Cakes	Jaffa Cakes
Kit Kat, Jaffa Cakes, Lays	Jaffa Cakes

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The Model

- We want to test the model of **utility maximization**
- Every object has a fixed utility value attached to it
- For example:
 - $U(\text{jaffa cakes})=10$
 - $U(\text{kit kat})=5$
 - $U(\text{lays})=2$
- In any choice set, choose the object with highest utility
 - For simplicity let's rule out ties – i.e utility function is one to one

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The Question

- As your PI I have asked you to go out and collect some choice data
- Is it consistent with the model of utility maximization?
- **Problem:** Our model contains 'unobservables'
 - We do not observe utilities
 - Kit Kats do not come with utility numbers stamped on them
 - Model says that people maximize utility, but as the experimenter I do not observe utility
- How can we proceed?

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Approach 1

- Pick a **particular** utility function
 - e.g. utility=calories
- Test whether this utility function can explain the data
 - e.g. Do people pick the option with the most calories?
- This is now a testable prediction
- And this is indeed how early economists proceeded
 - Bentham: Felicific Calculus
 - Proposed a classification of 12 pains and 14 pleasures, by which we might test the "happiness factor" of any action
- Problem?
- What does failure tell us?
 - Perhaps people do not maximize utility
 - Or perhaps utility is not equal to calories
 - Maybe Bentham overlooked a pleasure!

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Approach 2

- Ask the question: Is there **ANY** utility function that can explain the data?
 - i.e. we are agnostic about what utility is
- We require only that the person behaves as if they have some consistent utility function that they are using to make their choices

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Approach 2

- Note that this is what is sometimes referred to as 'as if' modelling
 - We don't observe utility directly
 - Only ask that we can find some utility function that explains choices
 - Subject behaves 'as if' they are maximizing utility
 - But they might be doing something completely different
- Using this approach, failure means that there is something wrong with the model of utility maximization
 - *Not* our definition of utility
- Equivalently, the predictions that come about from this method are exactly those you get from assuming utility maximization
 - Any additional predictions are coming from further assumptions you are making (such as functional forms)

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Aisha's Choices

Choice	Available Snacks	Chosen Snack
1	Jaffa Cakes, Kit Kat	Jaffa Cakes
2	Kit Kat, Lays	Kit Kat
3	Lays, Jaffa Cakes	Lays
4	Kit Kat, Jaffa Cakes, Lays	Jaffa Cakes

- Is there **any** utility function that can explain Aisha's choices
- No!
 - Choice 1 implies $u(\text{jaffa cake}) > u(\text{kit kat})$
 - Choice 2 implies $u(\text{kit kat}) > u(\text{lays})$
 - Choice 3 implies $u(\text{lays}) > u(\text{jaffa cakes})$
- Implies $u(\text{jaffa cake}) > u(\text{jaffa cake})$: Contradiction

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Brittney's Choices

Choice	Available Snacks	Chosen Snack
1	Jaffa Cakes, Kit Kat	Jaffa Cakes
2	Kit Kat, Lays	Kit Kat
3	Lays, Jaffa Cakes	Jaffa Cakes
4	Kit Kat, Jaffa Cakes, Lays	Kit Kat

- What about Brittney's Choices?
- No!
 - Choice 1 implies $u(\text{jaffa cake}) > u(\text{kit kat})$
 - Choice 4 implies $u(\text{kit kat}) > u(\text{jaffa cakes})$
- Contradiction

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Colvin's Choices

Choice	Available Snacks	Chosen Snack
1	Jaffa Cakes, Kit Kat	Jaffa Cakes
2	Kit Kat, Lays	Kit Kat
3	Lays, Jaffa Cakes	Jaffa Cakes
4	Kit Kat, Jaffa Cakes, Lays	Jaffa Cakes

- How about Colvin's Choices?
- Yes!
 - $u(\text{jaffa cakes}) > u(\text{kit kat}) > u(\text{lays})$
- Eg
 - $u(\text{jaffa cakes}) = 3$
 - $u(\text{kit kat}) = 2$
 - $U(\text{lays}) = 1$

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A General Rule

- Question: Is there a general rule that differentiates data sets that can be explained by some utility function from those that can't?

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A General Rule

- Question: Is there a general rule that differentiates data sets that can be explained by some utility function from those that can't?

The Independence of Irrelevant Alternatives

Say x is chosen from a set of alternatives A

B is a subset of A that contains x

Then x must be chosen from B

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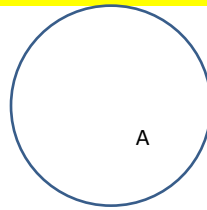
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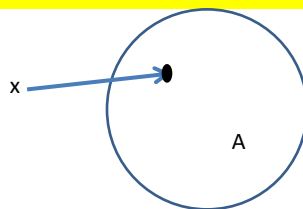
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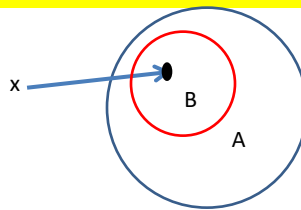
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Then x must be chosen from B



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Independence of Irrelevant Alternatives

Choice	Available Snacks	Chosen Snack
1	Jaffa Cakes, Kit Kat	
2	Kit Kat, Lays	
3	Lays, Jaffa Cakes	
4	Kit Kat, Jaffa Cakes, Lays	

- In our example, whatever is chosen in set 4 must always be chosen when it is available

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Aisha's Choices

Choice	Available Snacks	Chosen Snack
1	Jaffa Cakes, Kit Kat	Jaffa Cakes
2	Kit Kat, Lays	Kit Kat
3	Lays, Jaffa Cakes	Lays
4	Kit Kat, Jaffa Cakes, Lays	Jaffa Cakes

- Aisha's choices violate these condition
 - Jaffa cakes chosen in set 4
 - Lays chosen in set 3

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Brittney's Choices

Choice	Available Snacks	Chosen Snack
1	Jaffa Cakes, Kit Kat	Jaffa Cakes
2	Kit Kat, Lays	Kit Kat
3	Lays, Jaffa Cakes	Jaffa Cakes
4	Kit Kat, Jaffa Cakes, Lays	Kit Kat

- Also violated by Brittney's choices
 - Kit Kat chosen in set 4
 - Jaffa cakes chosen in set 1

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Colvin's Choices

Choice	Available Snacks	Chosen Snack
1	Jaffa Cakes, Kit Kat	Jaffa Cakes
2	Kit Kat, Lays	Kit Kat
3	Lays, Jaffa Cakes	Jaffa Cakes
4	Kit Kat, Jaffa Cakes, Lays	Jaffa Cakes

- Colvin's choices satisfy IIA
 - Jaffa cakes chosen in 4
 - Also chosen in 3 and 1

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A Necessary Condition

The Independence of Irrelevant Alternatives

Say x is chosen from a set of alternatives A

B is a subset of A that contains x

Then x must be chosen from B

- If we observe a utility maximizer, then they must satisfy IIA
 - If x is chosen from A , must have a higher utility than anything in A
 - B is a subset of A
 - x must have higher utility than anything in B
 - Should be chosen from B

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A Sufficient Condition?

The Independence of Irrelevant Alternatives

Say x is chosen from a set of alternatives A

B is a subset of A that contains x

Then x must be chosen from B

- Is it the case that, if IIA holds, there exists **some** utility function such that choices maximize utility **according to that utility function**?
- This would be great!
 - It means testing the condition is **the same as** testing the model of utility maximum
 - If the condition is satisfied then the person looks like a utility maximizer
 - If not, then they don't

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